STRATHMORE UNIVERSITY

School of Computing and Engineering Sciences

ICS 4111

Embedded Systems & IoT

PRACTICAL EXERCISE 3

Embedded System Design

Practical Demonstration of Ohm's Law

Group Members: **Group A: Daisies**	Samuel Gichuru (Group Leader) - **157660** Nicole Njeri - **146510** Alysa Macharia - **167016** Scott Muteti - **167997** Fatuma Marsa - **159056**
Course:	ICS 4111 - Embedded Systems & IoT
Semester:	April - July 2026
Date:	Monday, 25th May 2026
Venue:	SCES Makerspace Lab, Engineering Labs Building
Lecturer:	A. Vikiru

ICS 4111: Embedded Systems & IoT

Practical Exercise 3 - Group Meeting Minutes

Evidence of Group Participation

Meeting Details

Date:	Sunday, 25th May 2026
Time:	11:30 AM – 1:00 PM
Venue:	SCES Makerspace Lab, Engineering Labs Building, Strathmore University
Course:	ICS 4111- Embedded Systems & IoT
Exercise:	Practical Exercise 3: Embedded System Design
Minutes by:	Nicole Njeri

Attendance Register

#	Full Name	Role	Present
1	Samuel Gicuru	Group Leader / Arduino Programming / Circuit Assembly	**present**
2	Nicole Njeri	Circuit Assembly / Documentation	**present**
3	Alysa Macharia	Multimeter Measurements / Circuit Assembly	**present**
4	Fatuma Marsa	Circuit Assembly / Photography	**present**
5	Scott Muteti	PWM Setup / Photography	**present**

Agenda

#	Agenda Item	Owner	Duration
1	Setup and lab safety briefing	Samuel Gichuru	10 min
2	Experiment 1: Resistor colour code reading & multimeter resistance measurement	All members	20 min
3	Experiment 2: MCU + LED circuit, Blink sketch upload & PWM brightness control	All members	30 min
4	Experiment 3: Series resistor circuit assembly & voltage measurement	All members	15 min

5	Experiment 4: Parallel resistor circuit assembly & voltage measurement	All members	10 min
6	Documentation review and GitHub upload preparation	Nicole Njeri	5 min

Minutes of Proceedings

1. Setup : The group convened at the SCES Makerspace Lab. Samuel Gichuru led the safety briefing and distributed the components: one Arduino Uno microcontroller, two resistors, one LED, one breadboard, one potentiometer, and one multimeter per group.

2. Experiment 1 : Alysa Macharia identified the resistor colour bands and determined the nominal resistance value from the colour code. The group then used the multimeter to measure the actual resistance, with Scott Muteti holding the probes and Fatuma Marsa recording the reading. The multimeter displayed a measured value of 17.80 Ω.

3. Experiment 2 : Samuel Gichuru assembled the basic MCU + LED circuit on the breadboard following the EasyEDA schematic. He then uploaded the Blink sketch via Arduino IDE. Alysa Macharia used the multimeter to measure the circuit voltage, obtaining a reading of 2.156 V. Scott Muteti then reconfigured the circuit by adding the potentiometer for PWM control. The group documented the LED at three brightness levels: 25%, 50%, and 100%.

4. Experiment 3 : Samuel Gichuru and Fatuma Marsa assembled the series resistor circuit. The Blink sketch was re-uploaded by Nicole Njeri. Alysa Macharia measured the voltage across each resistor in the series configuration, recording a reading of 1.024V.

5. Experiment 4 : The group reconfigured the circuit into a parallel resistor configuration. Scott Muteti led the assembly while Nicole Njeri verified the connections against the schematic. Alysa Macharia measured the voltage, obtaining a reading of 2.08V.

6. Wrap-up : Nicole Njeri compiled all photographs and documentation. The group reviewed the deliverables against the exercise requirements. Files were organised for GitHub upload under the Embedded-Design branch of the Circuit-Design repository.

Experiment 1: Resistor Identification & Multimeter Measurement

Resistor Colour Code

The resistor was examined and its resistance value determined from its colour bands. The resistor has a blue body with three colour bands: Brown (1st band), Black (2nd band), and Black (3rd band / multiplier).

Using the 4-Band Colour Code Chart:

- 1st Band - Brown = 1
- 2nd Band -Black = 0
- 3rd Band (Multiplier) -Black = ×1 Ω
- Calculated Resistance = 10 × 1 = 10 Ω

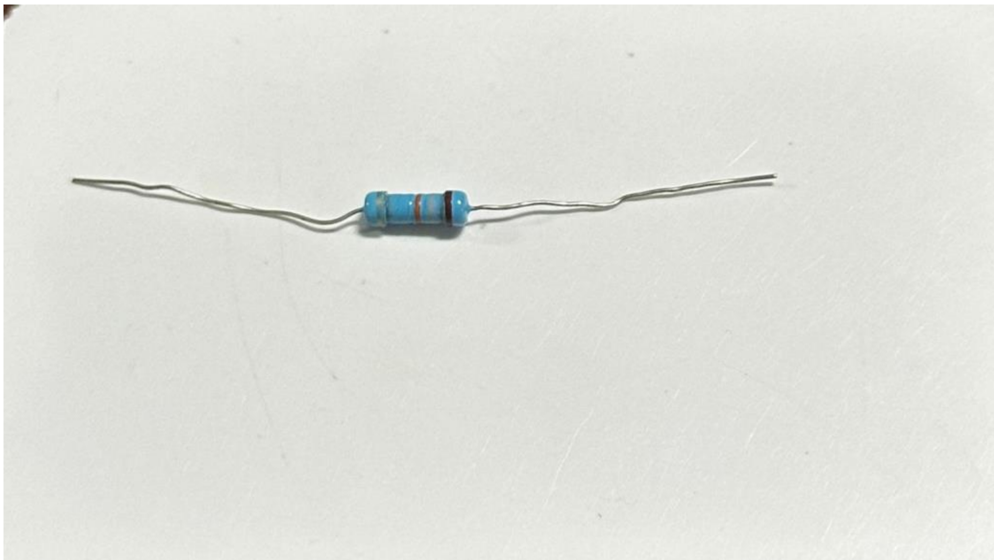

Figure 1: Resistor - Brown, Black, Black bands = 10 Ω nominal (Experiment 1)

Multimeter Resistance Measurement

A multimeter was used to measure the actual resistance of the resistor. The reading obtained was 17.80 Ω. This is higher than the nominal 10 Ω from the colour code due to probe contact resistance and lead resistance inherent in the multimeter setup.

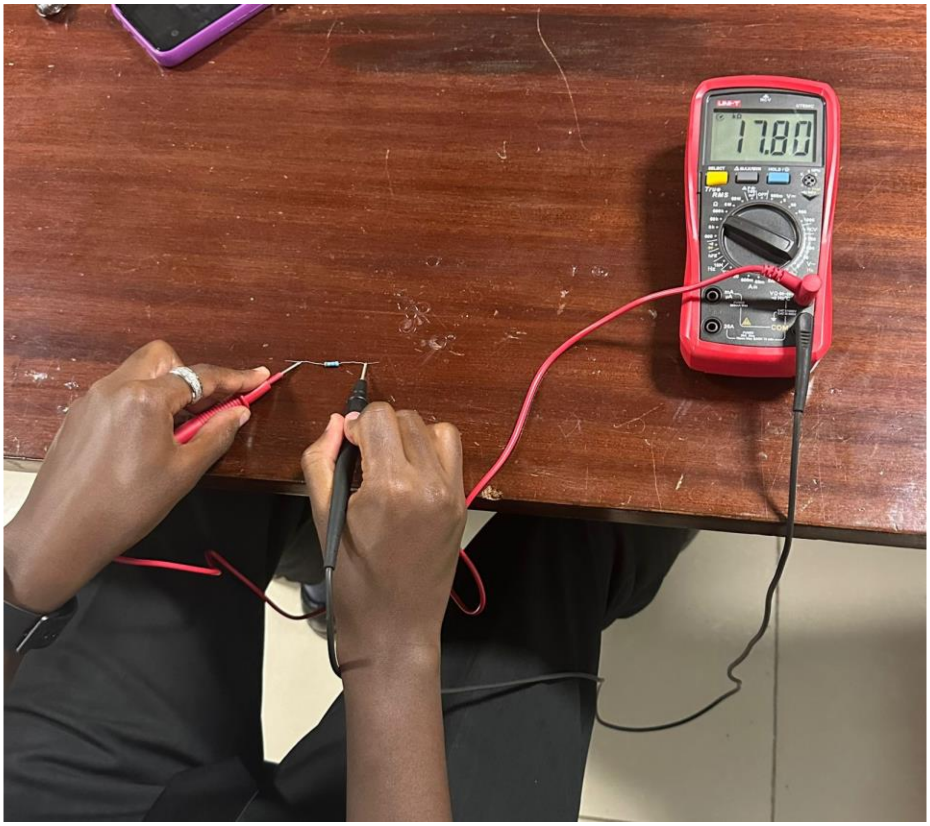

Figure 2: Multimeter reading - 17.80 Ω measured resistance (Experiment 1)

Why Do the Resistance Values Differ?

The colour code gives the nominal (theoretical) resistance value of 10 Ω, while the multimeter measured 17.80 Ω. The difference arises from several sources:

1. **Tolerance:** Resistors are manufactured within an accepted tolerance range. Without a visible gold or silver tolerance band on this resistor, the tolerance may be ±20%, meaning a 10 Ω resistor could legitimately read anywhere between 8 Ω and 12 Ω.

2. **Lead / Contact Resistance:** The multimeter probes and their leads have their own small resistance. When probing a low-resistance component, this added resistance becomes significant relative to the component being measured.

3. **Component Ageing:** Resistors can drift from their nominal value over time due to heat, humidity, and physical stress, particularly for carbon-film types.

4. **Measurement Technique:** Holding the probes loosely or at an angle increases contact resistance and inflates the reading.

In summary, the difference between the theoretical (10 Ω) and measured (17.80 Ω) values is primarily attributable to probe and lead resistance in the multimeter setup, combined with natural component tolerance.

Experiment 2: MCU + LED Circuit, Blink Sketch & PWM

Physical Implementation of MCU + LED Circuit

The Arduino Uno was connected to an LED on the breadboard, replicating the provided schematic.

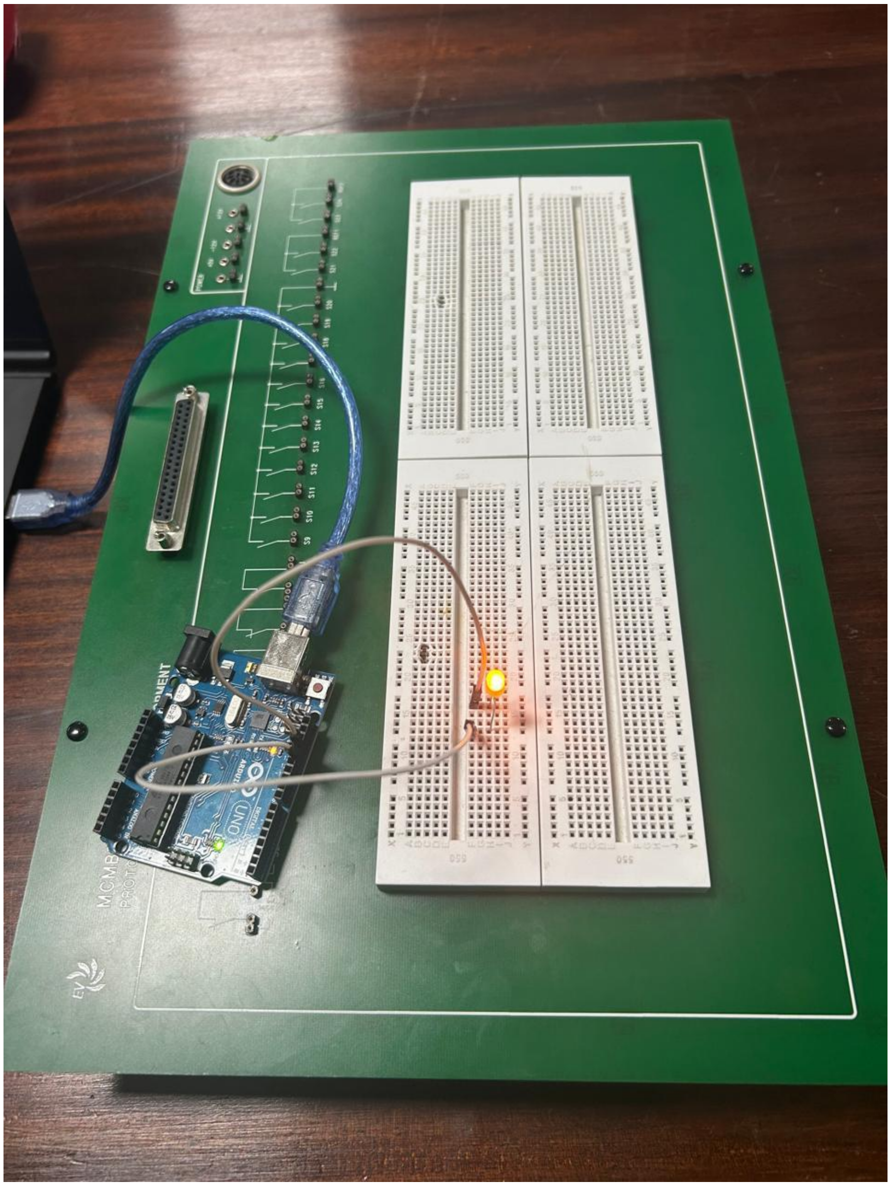

Figure 3: Physical implementation of the Arduino Uno + LED circuit (Experiment 2)

Deploying the Blink Sketch

Using the Arduino IDE, the built-in Blink example sketch was opened and uploaded to the Arduino Uno board via USB. The sketch configures the onboard LED pin as an output and alternates it HIGH and LOW with a 1000 ms (1 second) delay, causing the LED to blink on and off repeatedly. This confirms the microcontroller is correctly powered, the USB connection is functional, and the board is programmable.

Voltage Measurement Across the Circuit

A multimeter was used to measure the voltage passing through the MCU + LED circuit. Reading: 2.156 V.

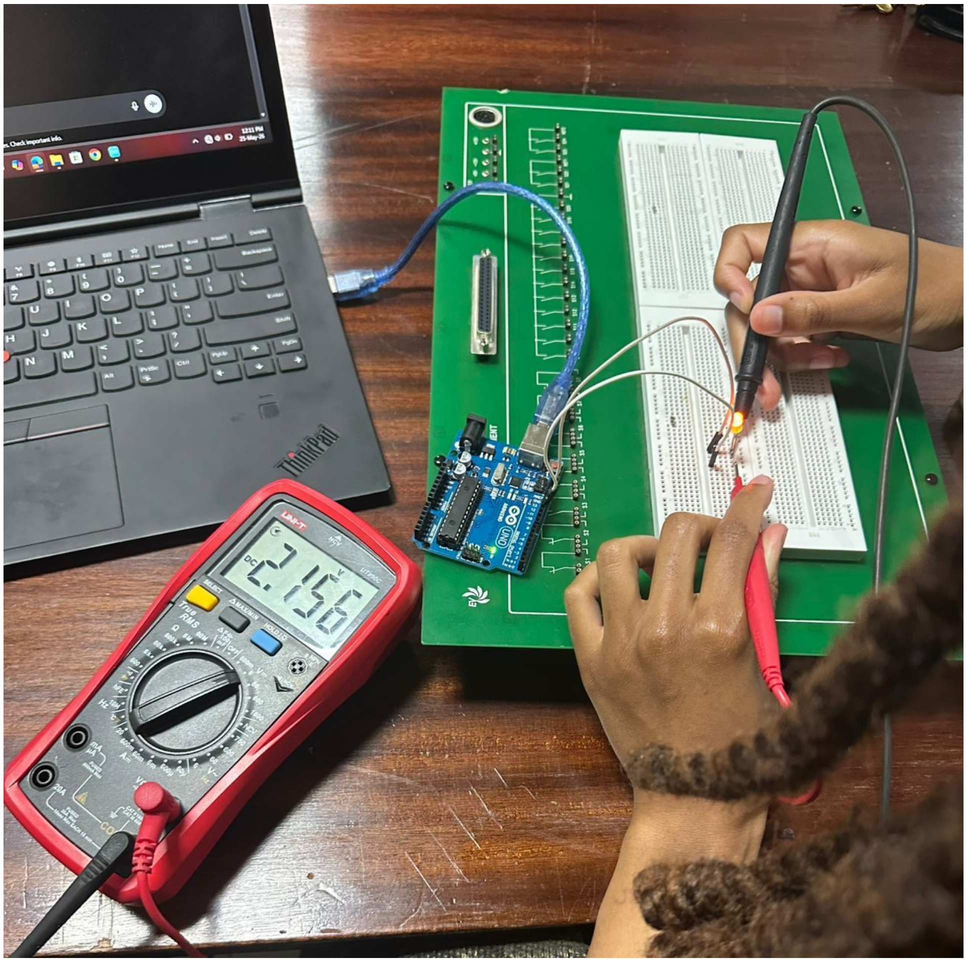

Figure 4: Multimeter reading- 2.156 V across the MCU + LED circuit (Experiment 2)

PWM Brightness Control Using a Potentiometer

A potentiometer was added to simulate PWM on the LED. Three brightness levels are documented below.

LED at 25% Brightness:

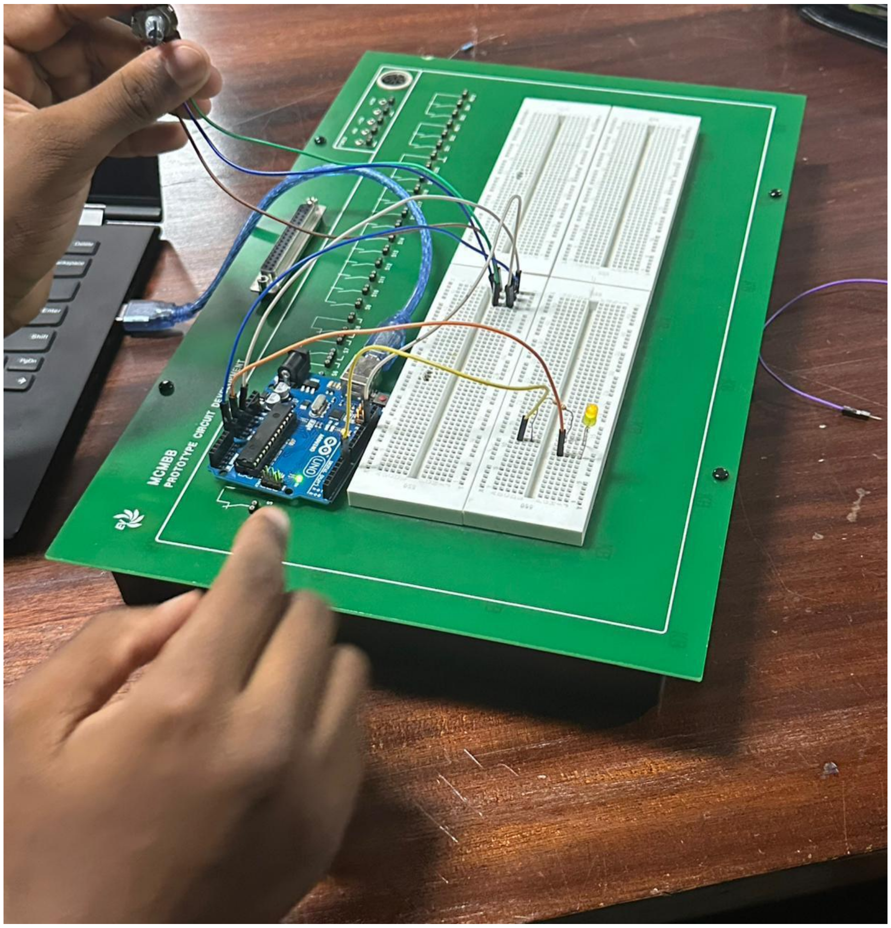

Figure 5: PWM circuit - LED at 25% brightness (Experiment 2)

LED at 50% Brightness:

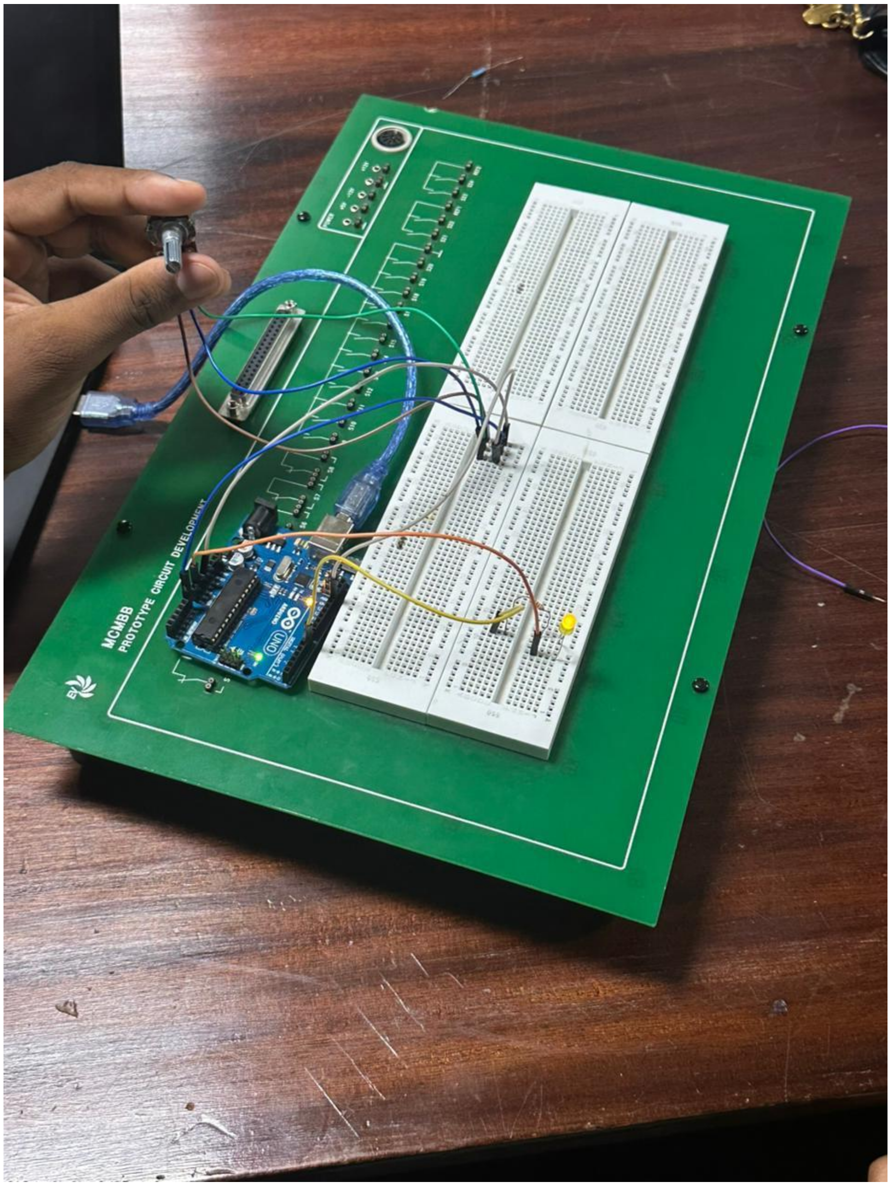

Figure 6: PWM circuit - LED at 50% brightness (Experiment 2)

LED at 100% Brightness:

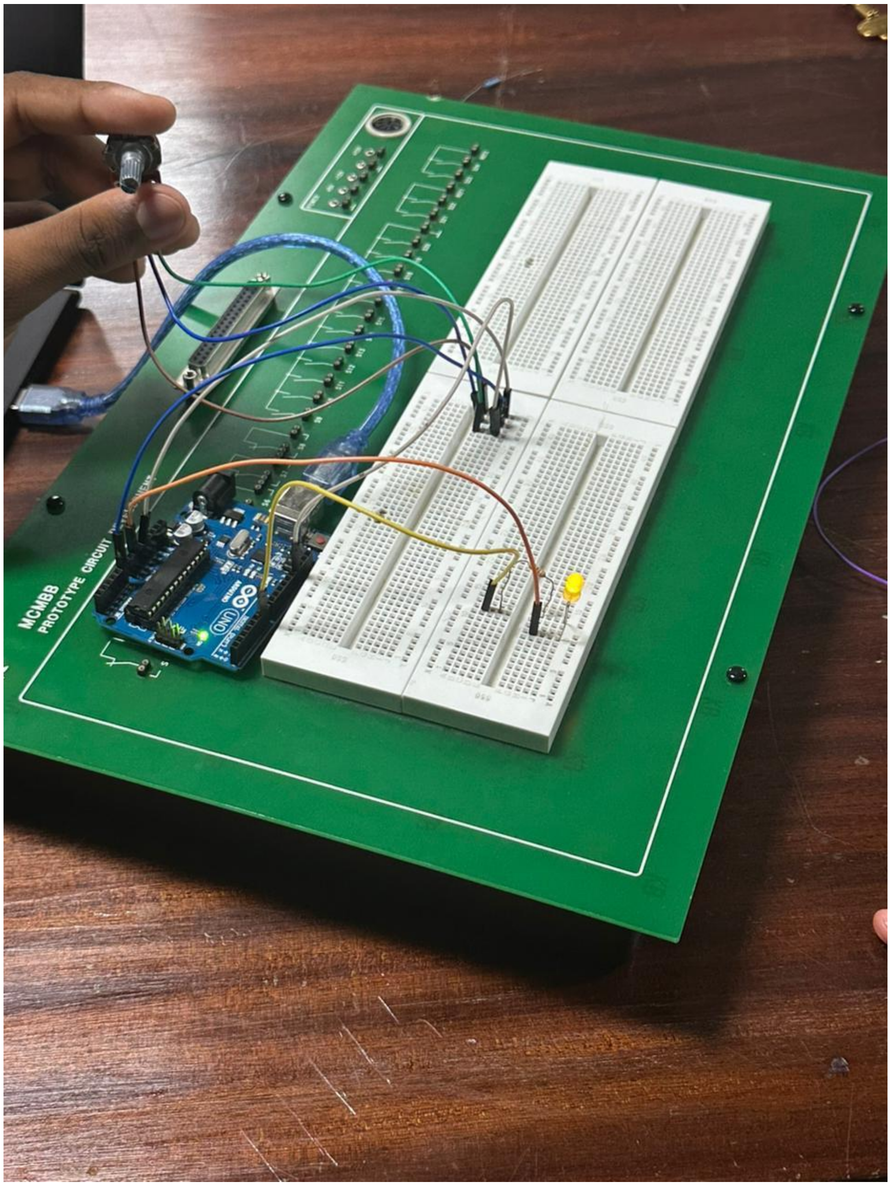

Figure 7: PWM circuit -LED at 100% brightness (Experiment 2)

Experiment 3: Resistors Connected in Series

Physical Implementation of Series Circuit

Both resistors were connected in series with the LED and Arduino Uno on the breadboard.

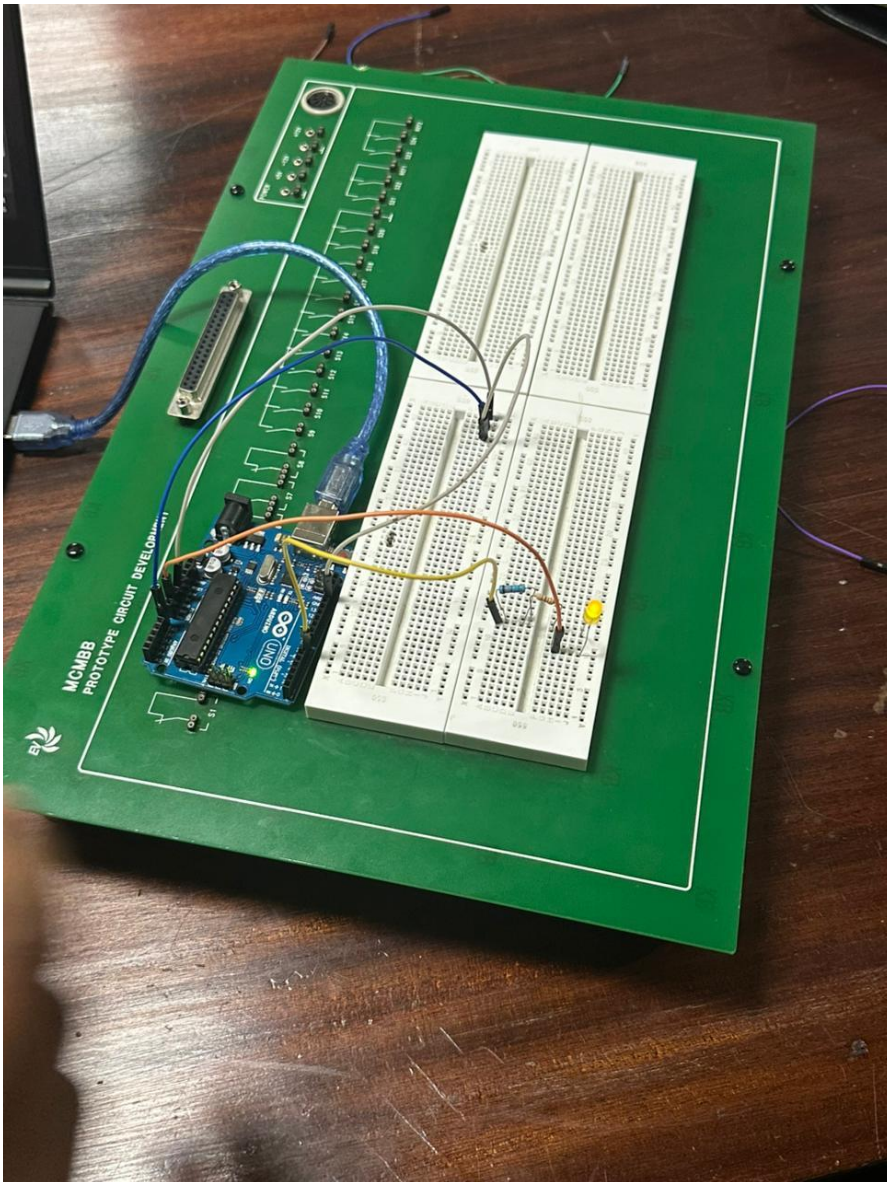

Figure 8: Physical implementation of the series resistor circuit (Experiment 3)

Voltage Measurement at Each Resistor (Series)

The multimeter was used to measure voltage across each resistor in the series configuration. Reading: 2.08 V.

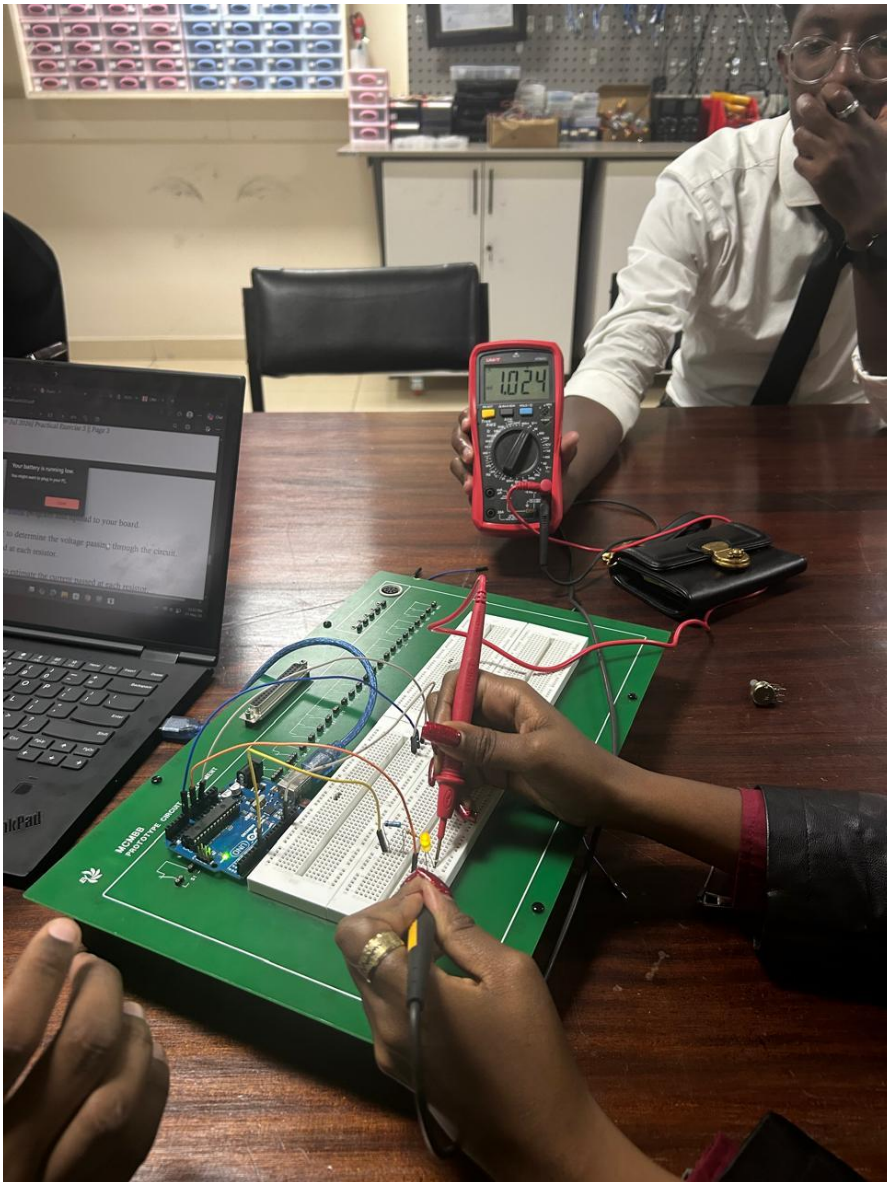

Figure 9: Multimeter reading – 1.024 V on the series circuit (Experiment 3)

Experiment Reflection - Ohm's Law: Current at Each Resistor (Series)

In a series circuit, the same current flows through every component. Using Ohm's Law **($I = V \div R$):**

Given values:

- **Total voltage measured across the circuit: $V_t = 1.024$ V**
- **Resistor 1 (R_1): 10 Ω (nominal from colour code)**
- **Resistor 2 (R_2): 10 Ω (nominal from colour code)**
- **Total series resistance: $R_t = R_1 + R_2 = 10 + 10 = 20$ Ω**

Estimated current through the circuit:

$I = V \div R_t = 1.024 \div 20 = 0.0512$ A

Since current is the same throughout a series circuit, the current through each resistor is:

$I_1 = I_2 = 0.0512$ A

Voltage Drop Across each resistor:

V1 = 0.0512 x 10 = 0.512V

V2 = 0.0512 x 10 = 0.512V

Experiment 4: Resistors Connected in Parallel

Physical Implementation of Parallel Circuit

Both resistors were connected in parallel with the LED and Arduino Uno on the breadboard.

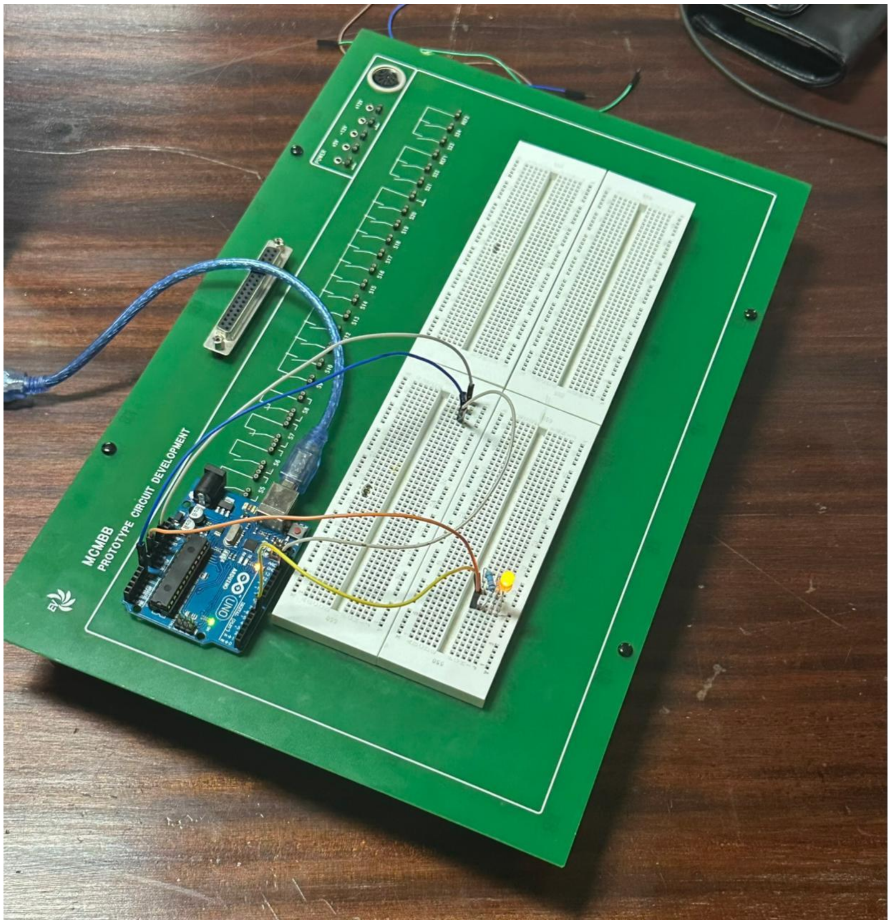

Figure 10: Physical implementation of the parallel resistor circuit (Experiment 4)

Voltage Measurement at Each Resistor (Parallel)

The multimeter was used to measure voltage across each resistor in the parallel configuration. Reading: 1.024 V.

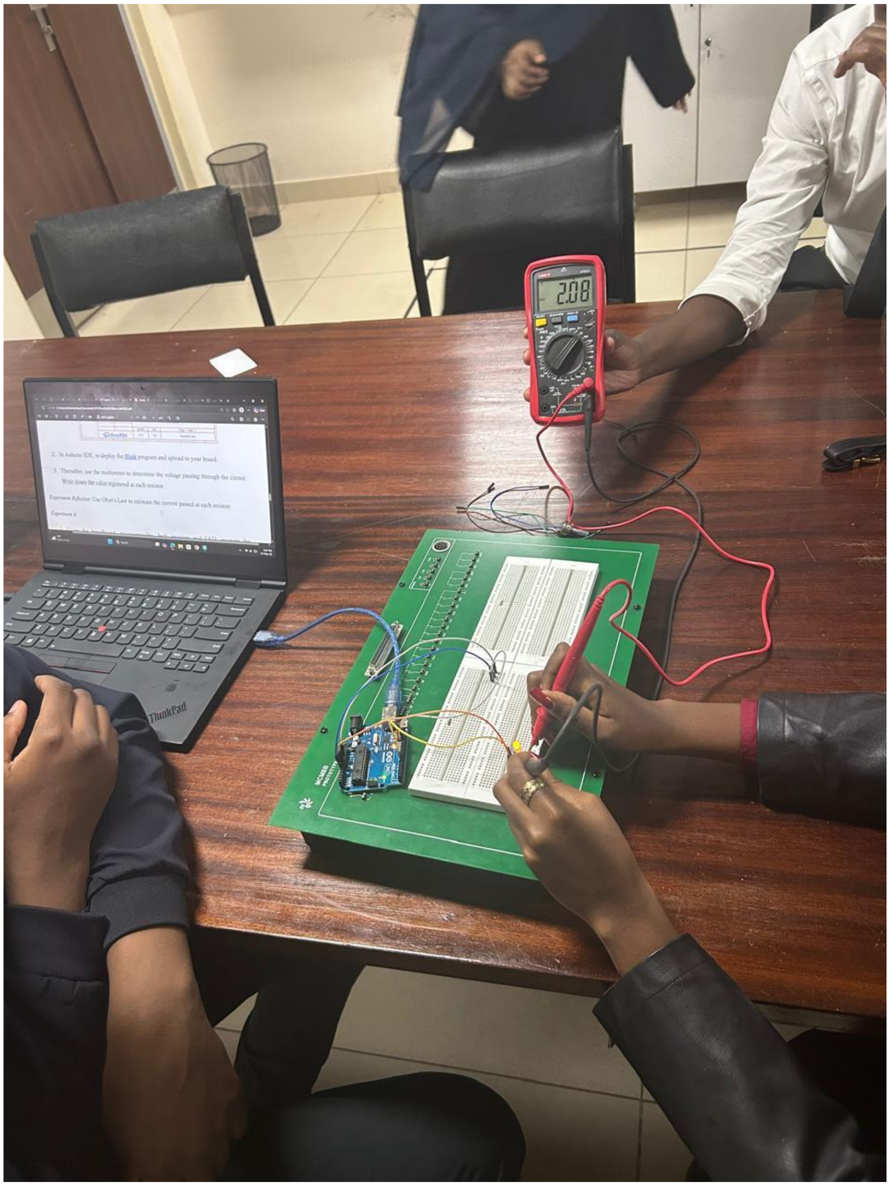

Figure 11: Multimeter reading- 2.08 V on the parallel circuit (Experiment 4)

Experiment Reflection - Ohm's Law: Current at Each Resistor (Parallel)

In a parallel circuit, the voltage across each branch is the same, but the current divides between branches. **Using Ohm's Law (I = V ÷ R):**

Given values:

- **Voltage measured across the parallel circuit: V = 2.08 V**
- **Resistor 1 (R_1): 10 Ω (nominal)**
- **Resistor 2 (R_2): 10 Ω (nominal)**

Equivalent parallel resistance:

$1/R = 1/R_1 + 1/R_2 = 1/10 + 1/10 = 0.2$

R(Total)= 5 Ω

Total Current:

$I = V/R$

$I = 2.08/5 = 0.416$ A

Current through each resistor:

In parallel, voltage is the same across each resistor, therefore;

$I1 = 2.08 / 10 = 0.208A$

$I2 = 2.08 / 10 = 0.208A$